Chemical Bonding

Cambridge IGCSE Chemistry 0620 · Topic 2.4–2.7 | EXTENDED = Supplement (Extended paper only) EDEXCEL = only if you sit Edexcel

The big idea

Atoms bond to get a full outer shell (the same electron arrangement as a noble gas). There are three ways to do it:

Bond	Between	What the electrons do	Structures it makes
Ionic	metal + non-metal	transferred from metal to non-metal	giant ionic lattice
Covalent	non-metal + non-metal	shared in pairs	simple molecules *or* giant covalent
Metallic	metal atoms	outer electrons delocalised (free to move)	giant metallic lattice

The structure (not just the bond) decides the properties. The summary table on the last page puts all four structures side by side.

1 Ions and ionic bonding

Definitions

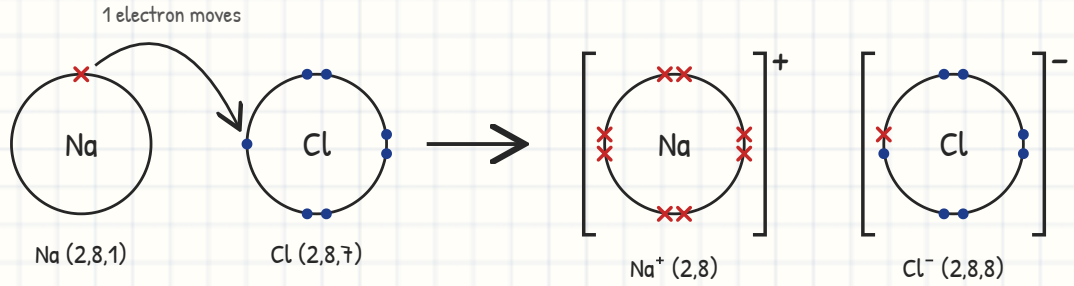
Ion – an atom or group of atoms with a charge, because it has lost or gained electrons.

Ionic bond – the strong electrostatic attraction between oppositely charged ions.

- Metals lose electrons → positive ions (cations)
- Non-metals gain electrons → negative ions (anions)
- The charge = number of electrons lost or gained to reach a full shell.

Group	I	II	III	V	VI	VII
Ion charge	+1	+2	+3	−3	−2	−1

Example: sodium + chlorine → sodium chloride



Dot-and-cross diagram. Na gives its 1 outer electron to Cl. Both ions now have a full outer shell. (Showing the Na^+ shell empty instead of with 8 × is also common — ask your teacher which they want.)

How to draw ionic dot-and-cross diagrams

1. Dots for one atom's electrons, crosses for the other's.
2. Outer shell only, unless the question asks for all shells.
3. Square brackets round each ion, charge outside at the top right.
4. Check: every ion has a full outer shell and the charges cancel out overall.

Any metal + non-metal EXTENDED

Magnesium oxide: Mg (2,8,2) gives 2 electrons to O (2,6) → Mg^{2+} (2,8) and O^{2-} (2,8).

Magnesium chloride: Mg gives 1 electron to *each* of two Cl atoms → Mg^{2+} and 2 Cl^- → $MgCl_2$.

Working out ionic formulas

The charges must add up to zero.

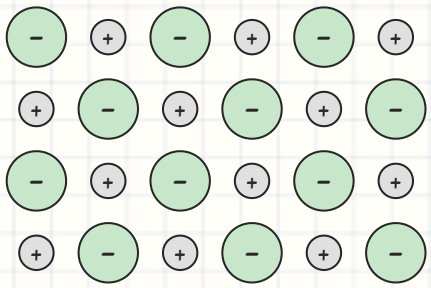
Na^+ and Cl^-	+1 −1 = 0	→ NaCl
Mg^{2+} and Cl^-	need 2 Cl^- to cancel +2	→ $MgCl_2$
Al^{3+} and O^{2-}	2 Al^{3+} = +6, 3 O^{2-} = −6	→ Al_2O_3

Shortcut: swap the charge numbers and drop the signs (Al^{3+} O^{2-} → Al_2O_3), then simplify if you can (Mg^{2+} O^{2-} → Mg_2O_2 → MgO).

Giant ionic lattice EXTENDED

A regular arrangement of alternating positive and negative ions, repeating in 3D in all directions.

There are no molecules. The formula NaCl just tells you the ratio of ions (1 : 1).



Sodium chloride lattice (one layer)

Properties of ionic compounds

Property	Why EXTENDED
High melting and boiling points	Giant lattice with lots of strong electrostatic attractions between oppositely charged ions. A lot of energy is needed to overcome them.
Conduct electricity when molten or dissolved in water	The ions are free to move and carry the charge.
Don't conduct when solid	The ions are held in fixed positions in the lattice, so they can't move.

Watch out

× "Electrons move" in a molten ionic compound → it's the ions that move.
× Calling NaCl a "molecule", or talking about "intermolecular forces" in ionic compounds.
× "Ionic bonds are weak" → they are strong.

2 Covalent bonds and simple molecules

Definition

Covalent bond – a pair of electrons shared between two atoms. It forms between non-metals, and each atom ends up with a full outer shell.

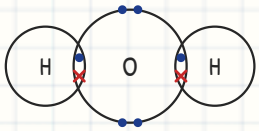
Exam wording

Cambridge: "a shared pair of electrons between two atoms, giving noble gas electronic configurations".
EDEXCEL "the strong electrostatic attraction between a shared pair of electrons and the nuclei of the bonded atoms".

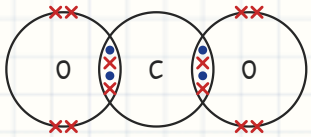
- Single bond = 1 shared pair (–)
- Double bond = 2 shared pairs (=)
- Triple bond = 3 shared pairs (≡)
- Electrons not used in bonds sit in pairs: lone pairs.

Number of bonds an atom makes = electrons it needs to fill its outer shell:

H → 1 Cl (Grp VII) → 1 O (Grp VI) → 2
N (Grp V) → 3 C (Grp IV) → 4



Water, H_2O: 2 single bonds, 2 lone pairs on O



Carbon dioxide, CO_2: 2 double bonds EXTENDED

How to draw covalent dot-and-cross diagrams

1. Overlap the outer shells of atoms that are bonded.
2. Put the shared electrons in the overlap: one dot + one cross for each bond.
3. Draw the leftover electrons in pairs (lone pairs).
4. Check: every atom has 8 in its outer shell (H has 2). Count each atom's own electrons too: O must have 6 of its own symbol, C 4, N 5, H 1.

Molecules you must be able to draw

Molecule	Displayed	Shared pairs	Lone pairs
Hydrogen, H_2	H–H	1	0
Chlorine, Cl_2	Cl–Cl	1	3 on each Cl
Hydrogen chloride, HCl	H–Cl	1	3 on Cl
Water, H_2O	H–O–H	2	2 on O
Ammonia, NH_3	N with 3 N–H	3	1 on N
Methane, CH_4	C with 4 C–H	4	0
EXTENDED also these			
Oxygen, O_2	O=O	2 (double)	2 on each O
Nitrogen, N_2	N≡N	3 (triple)	1 on each N
Carbon dioxide, CO_2	O=C=O	4 (2 doubles)	2 on each O
Ethene, C_2H_4	$H_2C=CH_2$	6 (1 double + 4 C–H)	0
Methanol, CH_3OH	$H_3C–O–H$	5	2 on O

Properties of simple molecular substances

Property	Why EXTENDED
Low melting and boiling points (often gases or liquids at room temperature)	There are only weak intermolecular forces between the molecules, so little energy is needed to overcome them. The covalent bonds *inside* the molecules are strong and do not break.
Don't conduct electricity	Molecules have no overall charge: there are no ions or free electrons to carry charge.

EDEXCEL Bigger molecules (higher relative molecular mass) → stronger intermolecular forces → higher boiling point.

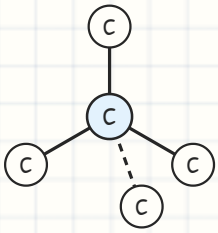
Watch out — the most common lost mark in this topic

× "When water boils, the covalent bonds break." → No. The intermolecular forces are overcome. Steam is still H_2O.

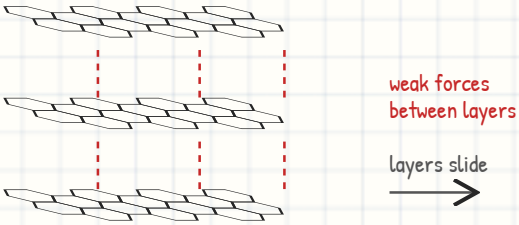
× "Covalent bonds are weak." → Covalent bonds are strong. The forces *between* molecules are weak.

3 Giant covalent structures

A huge network of atoms all joined by strong covalent bonds. No separate molecules. So they have very high melting points (you must break lots of strong covalent bonds), are insoluble, and don't conduct — except graphite.



Diamond: each C bonded to 4 others



Graphite: layers of hexagons, each C bonded to 3 others

	Diamond	Graphite
Bonds per C atom	4	3
Structure	rigid 3D network (tetrahedral)	flat layers of hexagons; weak forces between layers
Free electrons?	none — all 4 outer electrons used in bonds	1 delocalised electron per C atom
Hardness	very hard	soft and slippery
Conducts electricity?	no	yes
Melting point	very high	very high
Uses	cutting tools (drill tips, saw blades)	lubricant, electrodes (and pencils)

Link structure to use (common 2–3 mark question)

Diamond → cutting tools: each C is bonded to 4 others by strong covalent bonds in a rigid 3D structure, so it is very hard.

Graphite → lubricant: the layers are held together by weak forces, so they slide over each other easily.

Graphite → electrodes: it has delocalised electrons that can move through the structure, so it conducts electricity; it also has a very high melting point.

Silicon(IV) oxide, SiO_2 (sand) EXTENDED

Giant covalent: each Si is bonded to 4 O atoms and each O is bonded to 2 Si atoms (that 1 : 2 ratio is why the formula is SiO_2).

It is like diamond: hard, very high melting point, doesn't conduct (no free electrons or ions) — because both are giant networks of strong covalent bonds.

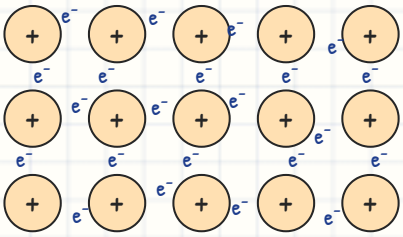
EDEXCEL C_{60} fullerene is a *simple molecule* (a ball of 60 C atoms): weak intermolecular forces, so a much lower melting point than diamond or graphite, and soft.

4 Metallic bonding EXTENDED

Definition

Metallic bonding – the electrostatic attraction between positive metal ions in a giant lattice and a "sea" of delocalised electrons.

Each metal atom lets go of its outer electrons. The electrons are shared by the whole structure and can move anywhere in it.



Positive ions in a sea of delocalised electrons

Property	Why
Good conductors of electricity (and heat)	The delocalised electrons can move through the structure and carry charge.
Malleable (hammered into shape) and ductile (drawn into wires)	The layers of positive ions can slide over each other. The sea of electrons keeps holding them together, so the metal changes shape instead of breaking.
Usually high melting points	Strong attraction between the positive ions and the delocalised electrons.

5 Putting it together

Structure	Examples	What's overcome when it melts	Melting point	Conducts as solid?	Conducts molten / dissolved?
Giant ionic	NaCl, MgO	strong attraction between oppositely charged ions	high	no	yes (ions move)
Simple molecular	H_2O, CO_2, CH_4, Cl_2	weak intermolecular forces	low	no	no
Giant covalent	diamond, graphite, SiO_2	many strong covalent bonds	very high	no (graphite yes)	– (insoluble)
Giant metallic	Cu, Fe, Al	strong attraction between positive ions and delocalised electrons	usually high	yes (electrons move)	yes

Identify the structure from data (classic exam question)

1. Conducts when solid? → yes: metal (or graphite)
2. If not – conducts when molten or dissolved? → yes: giant ionic
3. If not – look at the melting point: low → simple molecular; very high → giant covalent

Answer templates

"Explain why X has a high / low melting point" – 3 steps

1. Name the structure: "X has a giant ionic lattice / simple molecular structure / giant covalent structure / giant metallic lattice."
2. Name what must be overcome: "strong electrostatic attraction between oppositely charged ions" / "weak intermolecular forces" / "many strong covalent bonds" / "strong attraction between positive ions and delocalised electrons".
3. Energy: "so a lot of / only a little energy is needed to overcome them."

"Explain whether X conducts electricity"

Say which charged particles there are and whether they can move: "ions are free to move" / "delocalised electrons are free to move" / "there are no ions or free electrons, so nothing can carry the charge".

Self-check before you do the topic questions

- ☐ I can define ion, ionic bond, covalent bond and metallic bonding (metallic = Extended)
- ☐ I can draw the dot-and-cross diagram for NaCl (Extended: MgO, $MgCl_2$, any metal + non-metal)
- ☐ I can work out the formula of an ionic compound from the charges
- ☐ I can describe a giant ionic lattice EXTENDED
- ☐ I can draw H_2, Cl_2, HCl, H_2O, NH_3, CH_4 (Extended: O_2, N_2, CO_2, C_2H_4, CH_3OH)
- ☐ I can explain the melting point and conductivity of each of the 4 structures
- ☐ I can compare diamond and graphite and link each to its uses
- ☐ I can describe SiO_2 and say why it is like diamond EXTENDED
- ☐ I can identify a structure from melting point and conductivity data